

Mapping Superconductivity Research for AI: A Two-Axis Survey of 10,249 Preprints and Where Machine Learning Can Actually Help

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Abstract

PLACEHOLDER – written last.

Introduction

PLACEHOLDER – written last, after the body delivers.

Preliminaries and taxonomy

Terms this field overloads

A survey that spans nine material families and five modes of work has to fix its vocabulary first, because several of the terms below carry different meanings in different corners of the literature.

Critical temperature. T_c is quoted from several inequivalent operational definitions: the onset of a resistive drop, the midpoint of the transition, the zero-resistance point, and the temperature at which diamagnetic screening appears. The gap between onset and zero resistance is small in clean conventional samples and large in inhomogeneous ones, which is exactly where contested claims live. Throughout this survey we say which definition a reported number uses when the distinction bears on the claim, and we treat a resistive onset without a magnetic signature as weaker evidence than the two together.

Conventional and unconventional. We use *conventional* to mean pairing mediated by electron-phonon coupling and describable within Migdal-Eliashberg theory (Bardeen et al. 1957), and *unconventional* to mean a gap function that changes sign or breaks a lattice symmetry, whatever the glue. High-pressure hydrides are therefore conventional in mechanism while being extreme in T_c , and this survey groups them with the electron-phonon family rather than with the cuprates they are often headlined against.

Prediction versus discovery. A computed structure with a favourable calculated T_c is a prediction. A material that has been made and measured is a discovery. Papers routinely use “discovery” for the former, and the AI-for-materials literature does so more often than the superconductivity literature does. We reserve *discovery* for the experimental sense and say *predicted candidate* otherwise. This distinction carries most of the weight in Section 5.

AI-discovered. We treat a material as AI-discovered only when a machine-learning model selected or generated the candidate, the material was subsequently synthesised, and superconductivity was

measured. Weaker uses of the phrase – a screen that ranked a known compound highly, or a model that reproduced a T_c after the fact – are described as such.

The two axes

The taxonomy has two axes plus a lens (Figure 1).

Axis 1, material family, is what the paper is about: **conv** (conventional and electron-phonon systems, including high-pressure hydrides), **cuprate**, **febased**, **nickelate**, **unconv_other** (heavy fermion, ruthenate, organic, noncentrosymmetric bulk systems), **lowd** (two-dimensional, interface, moiré and kagome systems), **topo** (topological superconductivity and Majorana platforms), **device** (qubits, detectors, magnets, cables, circuits), and **general** (material-agnostic theory, formalism and instrumentation).

Axis 2, mode of inquiry, is how the work was done: **theory**, **abinitio**, **discovery**, **synthesis**, **characterization**. This axis deliberately does not encode whether a topic is applied. An earlier version of this taxonomy carried a sixth value, **application**, and it failed a measurement: it mixed *how* the work was done with *what it was for*, and characterization-versus-application was the single largest source of disagreement both for keyword rules and for two independently benchmarked language models. Application context is carried by the **device** family instead, so a qubit paper is labelled theory, synthesis or characterization like any other paper.

The AI lens applies only to papers that use a machine-learning method in their own work: **surrogate** (regression and classification models, symbolic regression), **gnn_potential** (graph networks and machine-learned interatomic potentials), **generative** (diffusion, VAE, flow models for structures and compositions), **llm** (language models, agents, literature mining), **nqs** (neural quantum states), and **autonomous_exp** (self-driving laboratories, closed-loop active learning).

The axes are the contribution of this survey, and Sections 3 through 5 are organised along them. Section 6 evaluates each cell for what AI can currently do in it.

Positioning relative to existing surveys

Two bodies of review literature already cover parts of this ground, and neither covers the join.

The first is per-family physics reviews. Recent examples treat nickelate superconductivity (Wang et al. 2025; Zhang et al. 2026) and the hydride programme. These are authoritative on mechanism and materials, and they say little about method infrastructure: what data exists, what is measurable at scale, where a model could be trained or tested.

The second is machine-learning-for-superconductors reviews. Recent examples survey materials-informatics approaches to superconductor discovery (Tran et al. 2025), machine-learning searches for new superconductors (Adiga and Waghmare 2025), AI-driven high-throughput screening (Gashmard et al. 2025), and language-model workflows for extraction and property prediction (Itani et al. 2025; Li et al. 2026). These are organised by method, and their material scope is dominated by the one task where tabulated data happens to exist – predicting T_c from composition.

This survey takes the join as its subject. It maps the whole field onto a family-by-mode grid, populates that grid from a systematically harvested corpus, and then asks of each cell what would have to be true for a machine-learning method to contribute there. The result is a statement about where the opportunities are *not*, which the method-organised reviews cannot make because they only look where the methods already are.

Corpus construction

Seed harvest. Every arXiv preprint listed under `cond-mat.supr-con`, as primary category or cross-list, submitted between 2021-09-01 and 2026-09-07, retrieved month by month through the arXiv API. This is a systematic, not a keyword, harvest: the selection criterion is the category label the authors themselves chose. It yields 10,303 records, of which 6,986 carry `cond-mat.supr-con` as primary category and 3,317 are cross-listed from elsewhere. Raw responses are cached per month so the harvest is reproducible.

Targeted supplement. The seed necessarily misses AI-for-materials work that never touches the superconductivity category. Two supplementary threads were harvested by keyword: AI applied to superconductivity from adjacent categories, and general AI-for-materials infrastructure needed for Section 6. These add 691 records tagged `supplement_t1` and `supplement_t2`. **The supplement is recency-biased by construction** – its queries are capped and sorted by submission date – so it is excluded from every growth statistic and from Table 1’s family-by-mode counts, where it appears only as a separate column.

Off-topic filtering. 745 records were flagged as not about superconductivity and excluded from all statistics. Within the systematically harvested seed the rate is 254 of 10,303, or **2.5%**, which is this corpus’s measured query-bleed. The rate is far higher in the supplement, as intended: the general AI-for-materials thread was designed to reach outside the field. After exclusion the working corpus is **10,249 records: 10,049 seed and 200 supplement**.

Labelling. Both axes and the AI lens were assigned by a language-model pass over title and abstract (`gemini-3.5-flash-lite`), after keyword rules were measured and found inadequate. Agreement with a hand-labelled 150-record stratified sample:

labels	family	mode of inquiry
keyword rules	80.1%	70.5%
language-model pass	85.6%	88.4%

Per-class recall for the model is 96% (theory), 92% (characterization), 92% (ab initio), 64% (synthesis) and 54% (discovery). The two small classes are the weak ones, and no claim in this survey rests on their counts alone. The AI lens was validated separately against 120 hand-labelled records, half drawn from each side of the rule-based tag: 88% precision and 81% recall on the AI-versus-not decision, 84% exact agreement on which method.

Annotator disclosure. All hand labels were produced by a single annotator. The figures above are therefore agreement between one human pass and one model pass, not accuracy against an external ground truth. Corpus, labels, prompts and scripts are released so that both passes can be re-run and re-scored.

Citation counts. Citation data comes from OpenAlex, resolved for 8,280 of the records. Over a five-year window raw counts favour older work mechanically, so where this survey ranks papers it ranks by citations per month since submission, and citation counts for work less than a year old are not read as impact at all.

Limitations. The corpus is arXiv-only, so it inherits arXiv’s coverage: strong in theory and condensed-matter experiment, weaker in applied-superconductivity engineering that publishes in IEEE venues, and blind to unpublished industrial work, which matters most for the device family.

Category self-assignment means a paper whose authors did not cross-list to `cond-mat.supr-con` is absent from the seed regardless of content. Labels are single-label where reality is often multi-label: a paper that grows a crystal and measures it gets one mode, chosen by which the abstract presents as the contribution.

Material families

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Modes of inquiry and their data infrastructure

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What AI has actually done in superconductivity

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An AI-readiness assessment

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Discussion

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Conclusion

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Data availability

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Adiga, Suhas, and Umesh V. Waghmare. 2025. “Accelerating the Search for Superconductors Using Machine Learning.” *Comput. Mater. Sci.*, 263, 114453 (2026), ahead of print. <https://doi.org/10.1016/j.commatsci.2025.114453>.

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